

Team 1 Design Document

NFR, FR, UR and KPI

Functional Requirements (**FR**) - Some of the functional requirements needed would be able to go click on and go through the other pages so they would be linked, be able to login, input their name, email, pathway, school ID to help get a meeting for the placement. Can identify the student and what they're doing

Non-functional Requirements (**NFR**) - The system should be able to handle as many students on the placement as possible using the system without slowing down and not taking long to update. Because it will be used for many students, their information and email that they put in need to also be encrypted so nobody else from outside gains access to it or leaks them

User Requirements (**UR**) - User requirements would be that students are able to sign up or register for updates. Can improve communication between other students by working well together on a project. Prevents scope so it doesn't go over budget, keep on adding to many features than planned that they could reasonably finish

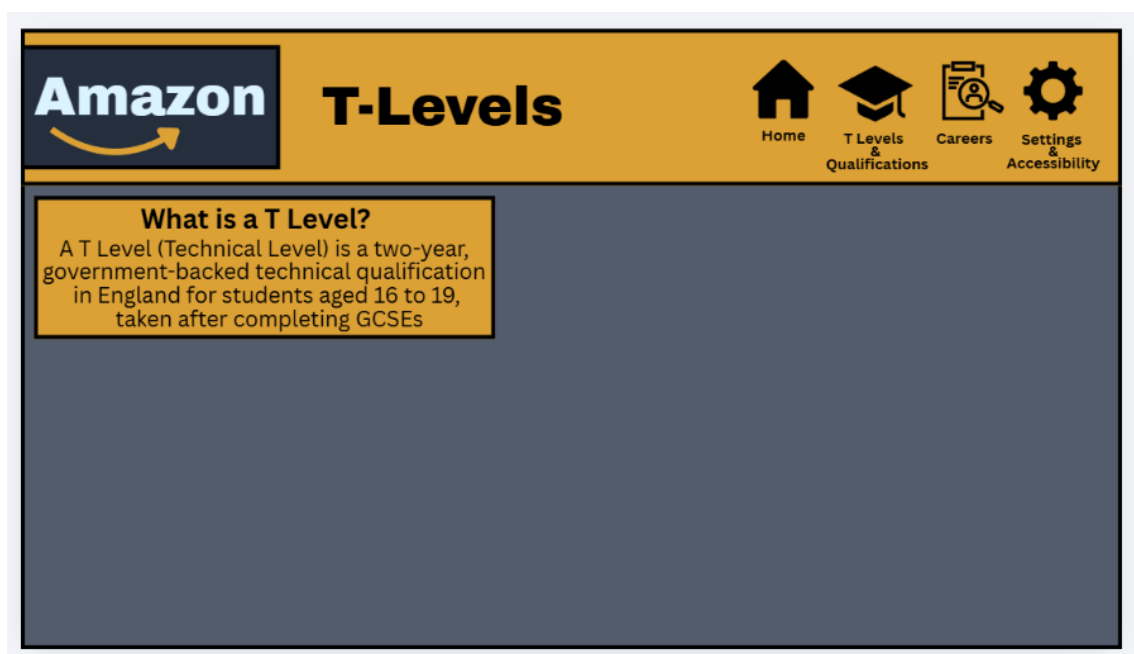
Key Performance Indicator (**KPI**) - number of student sign-ups per month. % of students complete the registration form fully (name, email, pathway, school id). Average page load time (should stay under 2-3 seconds). Number of successful placement meetings booked through the site. User drop-off rate (how many start the form but don't finish). Site uptime% (how often site is available without crashing).

Front End Design

Iteration 0:

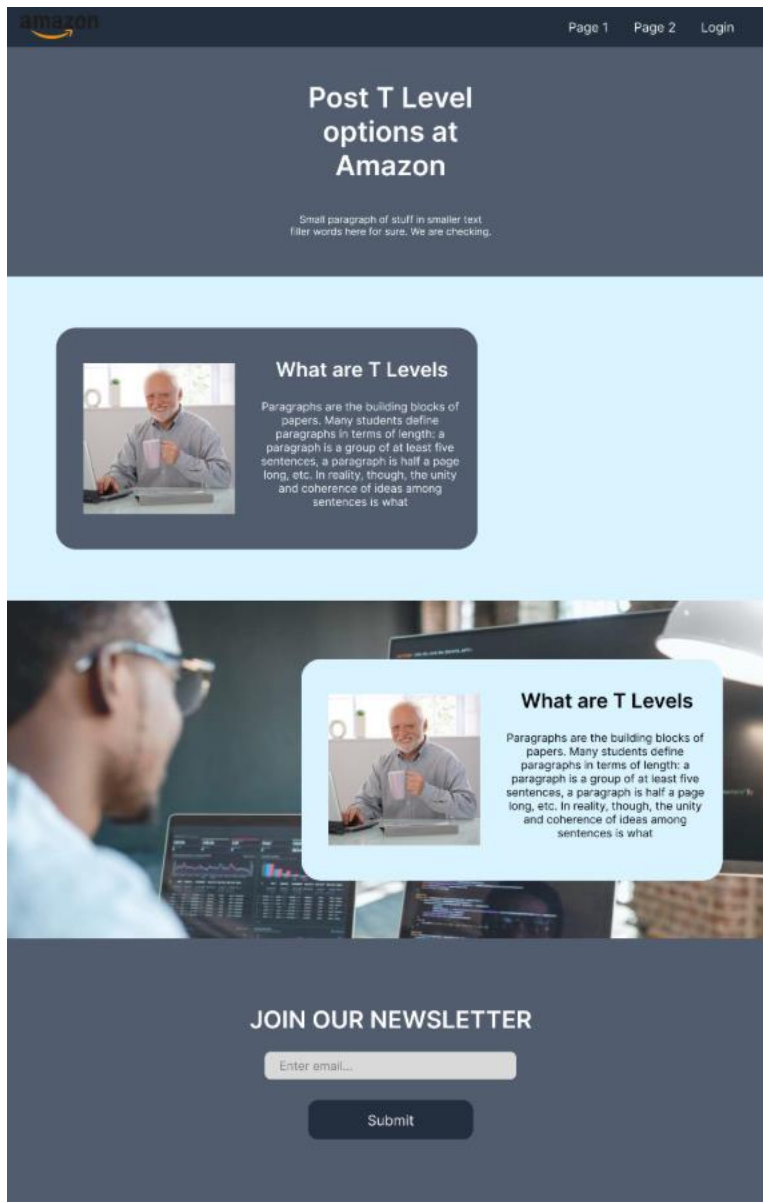
This is the first user interface design that we developed as a team. We tried to follow to a strike orange and dark blue colour scheme in order for it to

represent the normal standard for a Amazon designed product. A very basic design just having the groundwork for the layout, navbar, title and some basic information dotted in there to show the kind of thing we could design. We tried to fit everything into relative sizes and uniformed layouts so that everything will be accessible. The comments to take away here is that the layout is nice and everything is accessible, but the design isn't as great, and the information and layout still needs fleshing out. Hopefully the next design we can have more information areas and more fleshed out layout.



Iteration 1:

This iteration is a far different design but improved in certain ways while also lacking in other areas which makes it lack behind our later designs. Having a more design layout, with cantered titling and different sized fonts with structured paragraphs being the focus of this new design. The nav bar we decided would remain at the top with a change to have a smaller navbar compared to the one in the previous design. this design went for a more, dark blue focused design over the orange as we thought that amazon look was better than fully orange.



Iteration 2:

This design was another that was different to the one we used in the end. It had a lot of features that would be useful but didn't make it into the final design. Having a defined professional look, this page was built to be simple but bold and standing out with the fonts and image taking the majority of the user's attention when observing this page. It has a main white background which at first might seem like a wrong idea but it actually contrasts well with the text around the page. The fonts have different sizes to show difference between headings and actual paragraph texts. This allows the title to get the

users attention and hopefully make them interested enough to go on and read the actual page. The page also has some facts and statistics at the bottom which is another idea to catch users attention and get them hooked on the idea of choosing some of amazon's further pathways. Lastly with some buttons fitted for further pages with social features and sign up features. This design also tried to incorporate another page to it as we wanted to see the interaction and contrasts between the different pages we would build.

APPLICATIONS NOW OPEN FOR SEPTEMBER 2026

Launch your career with an Amazon T Level

Gain invaluable 20% hands-on industry placement experience at Amazon while completing 80% of your classroom studies. Designed for students aged 16-19.

Apply for Placements

Watch Program Overview

45 Days

Paid Experience

1-to-1

Industry Mentors

100%

Curriculum Support



THE QUALIFICATION

What is a T Level Placement at Amazon?

T Levels are a two-year qualification for 16 to 19-year-olds in England, designed with leading employers. They combine classroom learning with a substantial 45-day industry placement.



Equivalent to 3 A-Levels

It's better than being homeless or studying english language at Kingston uni



45 Days of Real Work

no larping



1-to-1 Mentor Network

Get paired with a dedicated Amazon professional who will support you through your career pathway, portfolio, and personal growth.

Iteration 3: Yousuf

Using the prior designs , we created our third iteration. This design took the previous iterations and designs we had to develop and improve upon our existing first design. This now has quite a lot of features added to it with more white background for text contrast. Good colouring schemes and buttons for downloadable content and copy link functions. For this design we have added a lot more information onto the page to flesh it out a lot more and added imaging to brighten up the webpage overall. This design also incorporated other parts of the website that would be made like sign up page for the newsletter however these are very bland and undesign3ed. While the bland design allows for a clear set out goal of the pages function there is the issue of it being boring and ugly.

Page 1Page 2Login

Post T Level options at Amazon

Small paragraph of stuff in smaller text that reads more like a paragraph, like the one you're reading.

T Level, what's a T Level

Paragraphs are the building blocks of papers. Many students define paragraphs in terms of length: a paragraph is a group of at least five sentences, a paragraph is half a page long, etc. In reality, though, the unity and coherence of ideas among sentences is what

45 Years

Of Something

200 Liters

Of Something Else

1-to-1

Coaching

Download PDF



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JOIN OUR NEWSLETTER

Submit

Page 1Page 2Login

Signup

Username

Enter username here...

Email

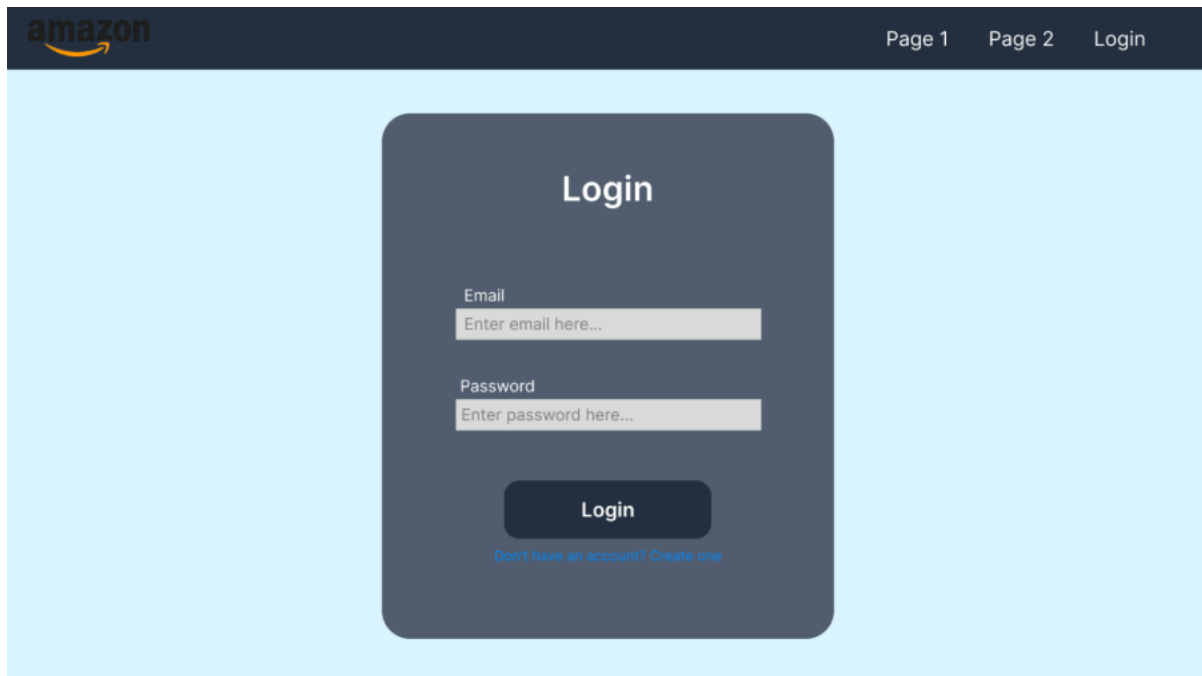
Enter email here...

Password

Enter password here...

Signup

[Already have an account? Login](#)



Other pages should just be a copy of this design but altered with the text and images.

Iteration 4:

In this design we have taken all other designs and ideas and combined them into one design for us to use for our development. This design adds more detailed images and backgrounds while also having a clear title area, nav bar and footer that includes links to other amazon websites. This design is by far our best design being the one we will develop as our final product. This has features like downloadable features, copy link functions, statistics and facts, and labelled and sectioned out clear paragraphs of information to give to the user with clear headings. This design also has a better design for our sign up newsletter section that was bland before but now fully made and designed tom give the user a nicer experience.

Post T Level options at Amazon

Discover how your technical T Level qualification can launch a high-impact engineering career or funded degree apprenticeship at Amazon.

amazon

Homepage Page 1 Page 2 Newsletter

Advanced Apprenticeships

Take ownership of meaningful engineering projects that directly impact the daily experiences of millions of global customers. From optimizing AWS cloud infrastructure for peak efficiency to coding and deploying innovative platform features, your fresh perspective will help solve complex problems and genuinely shape the future of our technology

3-4 Years
Program Duration

£30,000
Starting Salary

100%
Tuition Funded

[Download PDF](#)



Expert 1-to-1 Mentorship

Never navigate your career alone. We pair entry-level talent with dedicated senior engineers to provide continuous feedback, teach industry best practices, and help you confidently build your technical capabilities.

85%+
Graduation Rate

86%
Stay at Amazon

7,000+
UK Opportunities

Engineering at Scale

Take ownership of optimizing AWS cloud infrastructure for peak efficiency to coding and deploying innovative platform features, your fresh perspective will help solve complex problems and genuinely shape the future of our technology

50+
Different Schemes

50%+
STEAM Careers

1,000+
New Roles Yearly



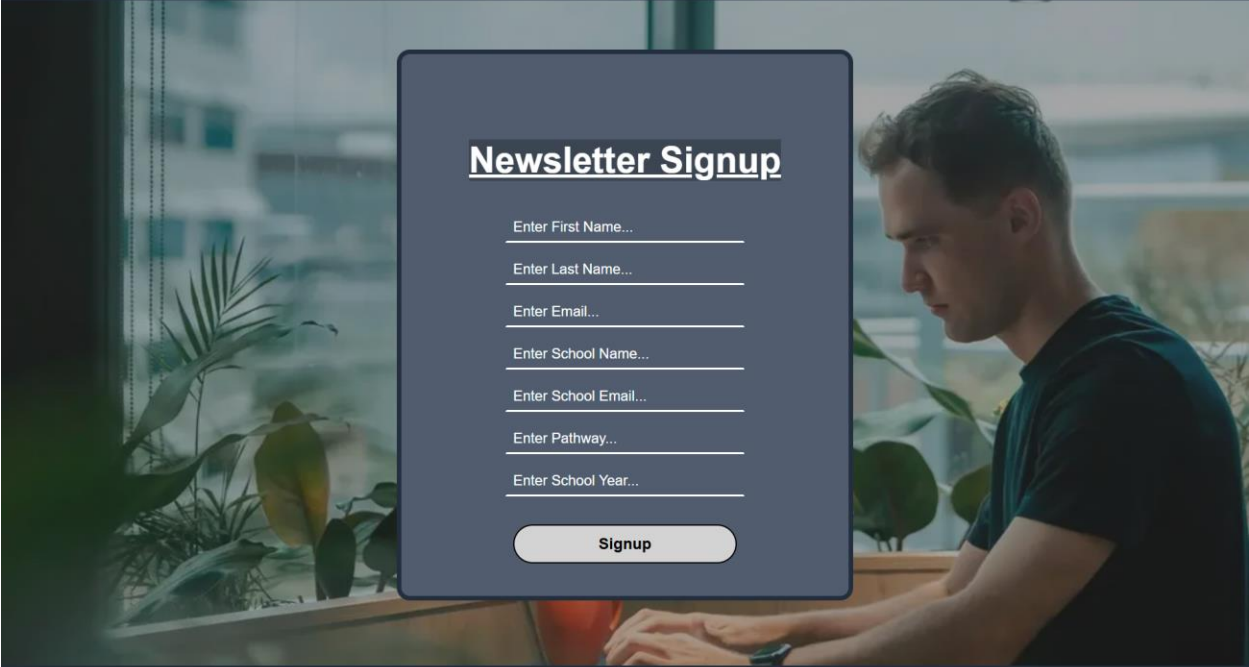
amazon

[Amazon Careers](#)

[Amazon](#)

[Amazon News](#)

 [Homepage](#) [Page 1](#) [Page 2](#) [Newsletter](#)



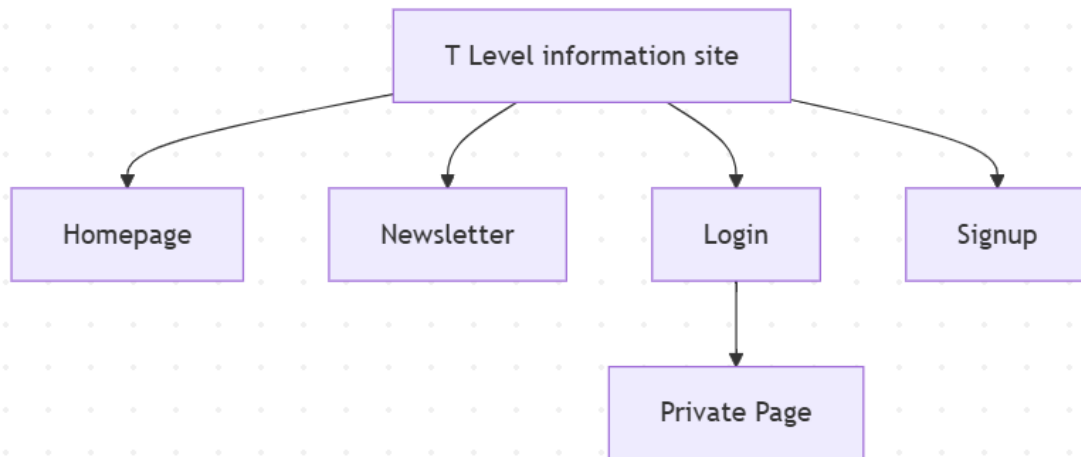
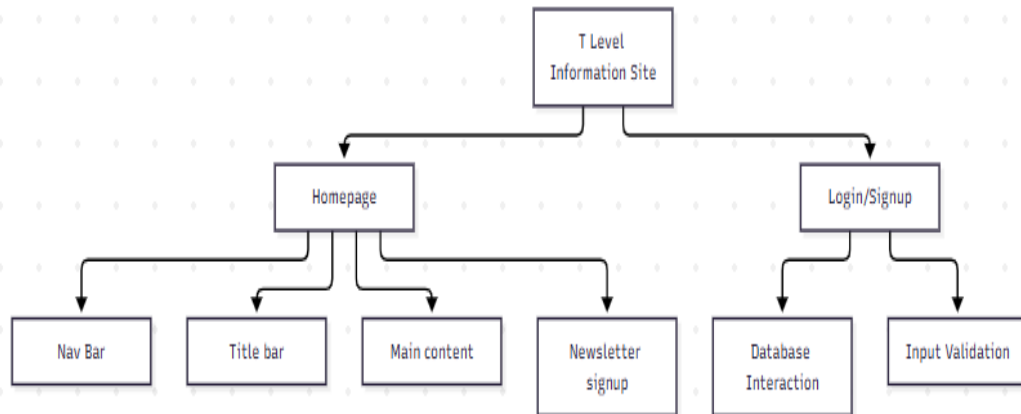
Newsletter Signup

Signup



[Amazon Careers](#) [Amazon](#) [Amazon News](#)

Overview / decomposition diagram1



Colour Scheme

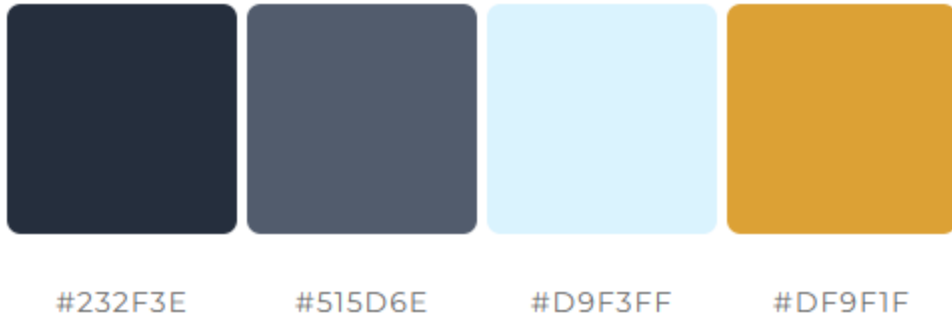
#232f3e

#515d6e

#d9f3ff

#df9f1f

Spot Palette

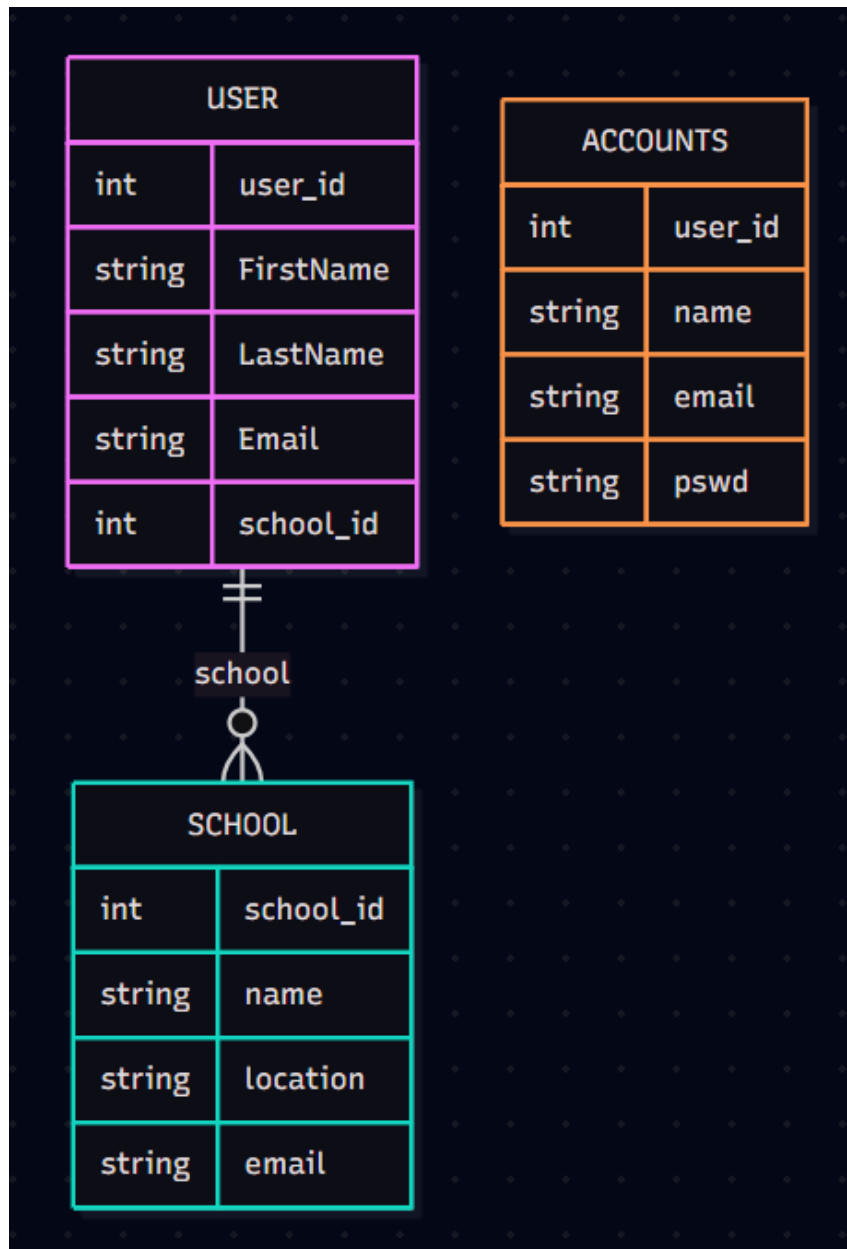


Data Dictionary

Due to the scale of the project, it did not seem worthwhile adding more complexity to the database and the database interaction. This is since the storage savings will not be significant, whereas the time loss used to set up a multi table would be significant in the timescale given.

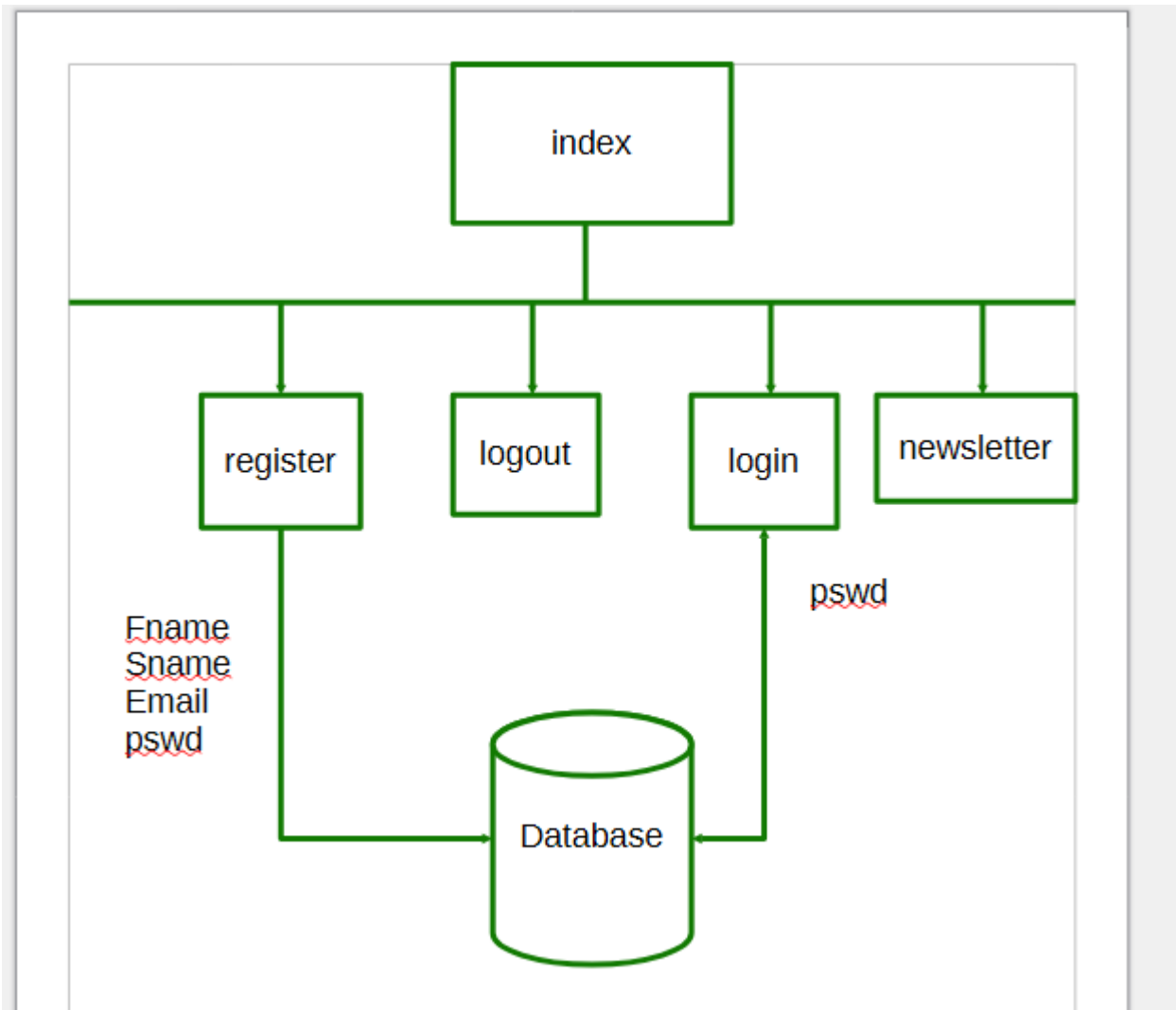
Data Base Design

We have chosen the inefficient data base this is where we are getting a single table because the scale of the project is too small to make it complicated. We have done this mainly because we don't want to waste time on the more complex and more difficult data base. This would also make the data base less cluttered and easier to understand.



Data Flow Diagram

Overview of all pages on our website. Represent database with cylinder. one way data link from database. This shows that you only use two tables that are linked by the primary and foreign key `school_id`. This is good that we are using this type of data link because it is simpler and easier to code out and implement into our database and into our website. We also think that it will save time and resources



Data Structures (Variable List)

String pswd
String fname
String lname
String school email
String email
String school
Int userid

Algorithm Designs (Pseudo code)

```
handleSignup
```

```
pswd = $POST['pswd']  
if pswdValid(pswd):  
    updateDatabase($POST)  
    return "Signup Successful"  
else:  
    return "Invalid Details Given"
```

```
pswdValid pswd
```

```
if pswd.len < 8:  
    return False  
elif specialcharacter not in pswd:  
    return False  
return True
```

Test Strategy

The first type of testing I would do is functional testing. This is when you are given the “finished” code, and try to find different ways to break it, by actually using it, to identify flaws.

Identify Functional Requirements: Understand the expected behavior of the application.

Develop a Testing Strategy: Allocate resources and define schedules.

Create Test Cases: Design test cases and test data based on requirements.

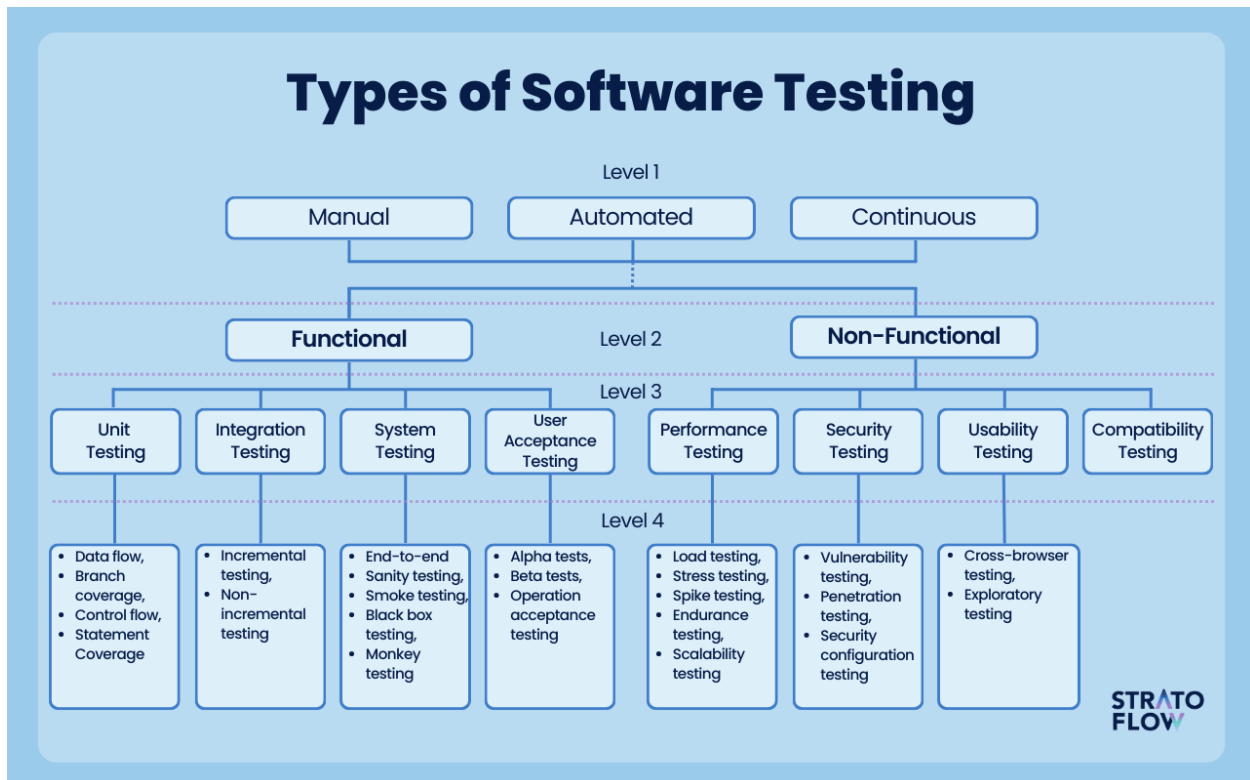
Set Up Test Environment: Configure necessary hardware, software, and testing tools.

Execute Test Cases: Run test cases and compare results with expected outcomes.

Log Defects: Report identified defects with details for resolution.

Re-test: Re-execute failed test cases after defects are fixed.

Document Results: Review testing activities and prepare closure reports.



Usability testing

We could also do the usability testing this is when we will see how the user-friendly features are displayed. We would do this by showing the website to a friend and seeing how they would perceive the website and its main image. Basically, you should just give real users just a survey and expect feedback.

Security Testing

We would use Vulnerability testing so we could see if there is any venerable code and see if there is any way people can find an exploit and find it before the big release. This is because if there was a big data leak our company could be getting a lawsuit or becoming bankrupt.

Performance Testing

We would see if there were any performance reductions with the website and if there are any lag spikes because of constant traffic and we would need to see if there was a breach/infiltration how well would our systems could withstand a DDOS attack and if the network could prevent it from shutting down. For example, we would make a lot of people could sign up or visit the site at the same time and then we would also see if there was

Compatibility testing

Is when a program can be run on different browsers and still be presented as the same format we would also need to find out if certain operations systems can run the right formats and if you had to run it on a phone or a mobile device would it be displayed properly.

Integration testing

Integration testing is a software testing approach in which various application components or modules are joined and tested to evaluate how well they work together. Integration testing seeks to help ensure that these assembled parts can communicate with each other and interact successfully.

Unit Testing

Unit testing is when you isolate one thing and see how it can be improved or fixed. We would do this to find a piece of the code that doesn't work and then either fix it or improve it.

Erroneous data:

Inputting wrong data to see how good the error handling is, or whether it just crashes

Boundary data:

For example, if a program check if a value if between 1 and 10, inputting 1 or 10 would be classed as boundary data as it is “on the boundary of the values”

Valid data:

Checking if any of the data is correct and meets quality requirements before executing its capabilities

Invalid data:

Checking if any of the data is incorrect and if it is then it should be corrected so the system can be executed properly.

<u>Date Of Test</u>	<u>Component To Be Tested</u>	<u>Type of Test to be Carried Out</u>	<u>Prerequisites and Dependencies</u>
17/09/2026	We would be testing the how much load can the servers and website take without crashing or the database being overloaded this is with the sign-up pages	The type of testing we would use is performance testing this is when we send a lot of traffic to the database and see if the servers could withstand it	
17/09/2026	We would be testing if the system could handle erroneous data	We would try doing security testing by inputting erroneous data into the website to see if the website ends up throwing any errors	
17/09/2026	We would be testing the different pages of the site	Functional testing – we would test the different functions of the login and sign-up pages, such as creating an account, logging in and logging out, to make sure they work correctly	
17/09/2026	Making sure we can use different buttons to go to other pages of the site such as going back to amazon shopping	Usability testing – we would test how easy the website is to use by checking that users can easily find pages, understand the buttons and	

		navigate around the website.	
17/09/2026	Making sure the site works on different browsers or devices	Compatibility testing – we would test the website on different browsers and devices, such as Chrome, Edge, phones and computers, to make sure it works correctly on each one.	
21/09/2026	Making different user accounts to see if the system can handle it	Functional testing – we would be testing the PHP system to make sure that it can handle different accounts being added to the database	
21/09/2026	Having different data be entered into the system to see if the system can handle user error from data entry.	Functional testing – we would be testing the website system to check if it can handle users incorrect data.	